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## 3 Ways to Transfer Files Between a Virtual Machine and PC on VMware and VirtualBox

Need to share data from a host PC to a virtual machine OS? Learn how to share data using copy and paste, USB, and shared folders.

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Virtual machines allow you to run a different operating system in an isolated environment. Because virtual machines, by nature, are isolated, you cannot directly access files on your host PC.





Copying text or files from the host PC to the guest, or vice-versa, is often a major sticking point for virtual machine users. Although it may seem complicated, the actual process to move files from VM to host and back again is straightforward.

A video player interface. The video area is black with a white circular arrow icon and the text 'Replay' next to it. Below that is a white globe icon and the text 'Learn More'. Below the video area, the text 'Meet Our Solvers: Ishita Agrawal' is displayed. Below that, it says 'SPONSORED BY PWC INDIA' and there is an orange 'Learn More' button.

Let's look at the different methods for transferring files between your virtual machine guest operating system and host PC using VirtualBox or VMware.

## How To Transfer Files From Virtual Machine To Host

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A virtual machine is a software environment that emulates the hardware required to install an operating system (OS). In very general terms, this lets you install an operating system on an existing OS, much like you would an app.

The options for sharing data between a host PC and guest operating system running in a virtual machine are what you might expect:

- Drag and drop (also known as copy and paste)
- USB drive
- Shared Folder

Each of these methods is ideally suited for a specific type of data and expected use. For example, copy and paste is best for sharing text and small files, such as copying code from a browser on your host PC into a terminal session in the guest OS.

Below, we'll look at sharing data using these three methods on virtual machines running on Oracle VirtualBox and VMware Workstation Player.

### 1. Drag & Drop and Shared Clipboard

The simplest option to transfer files between your host PC and a virtual machine is to set up a shared clipboard and/or drag and drop. This will allow you to use a common clipboard, meaning you will be free to copy/paste text/images and files between the virtual machine and your host PC.

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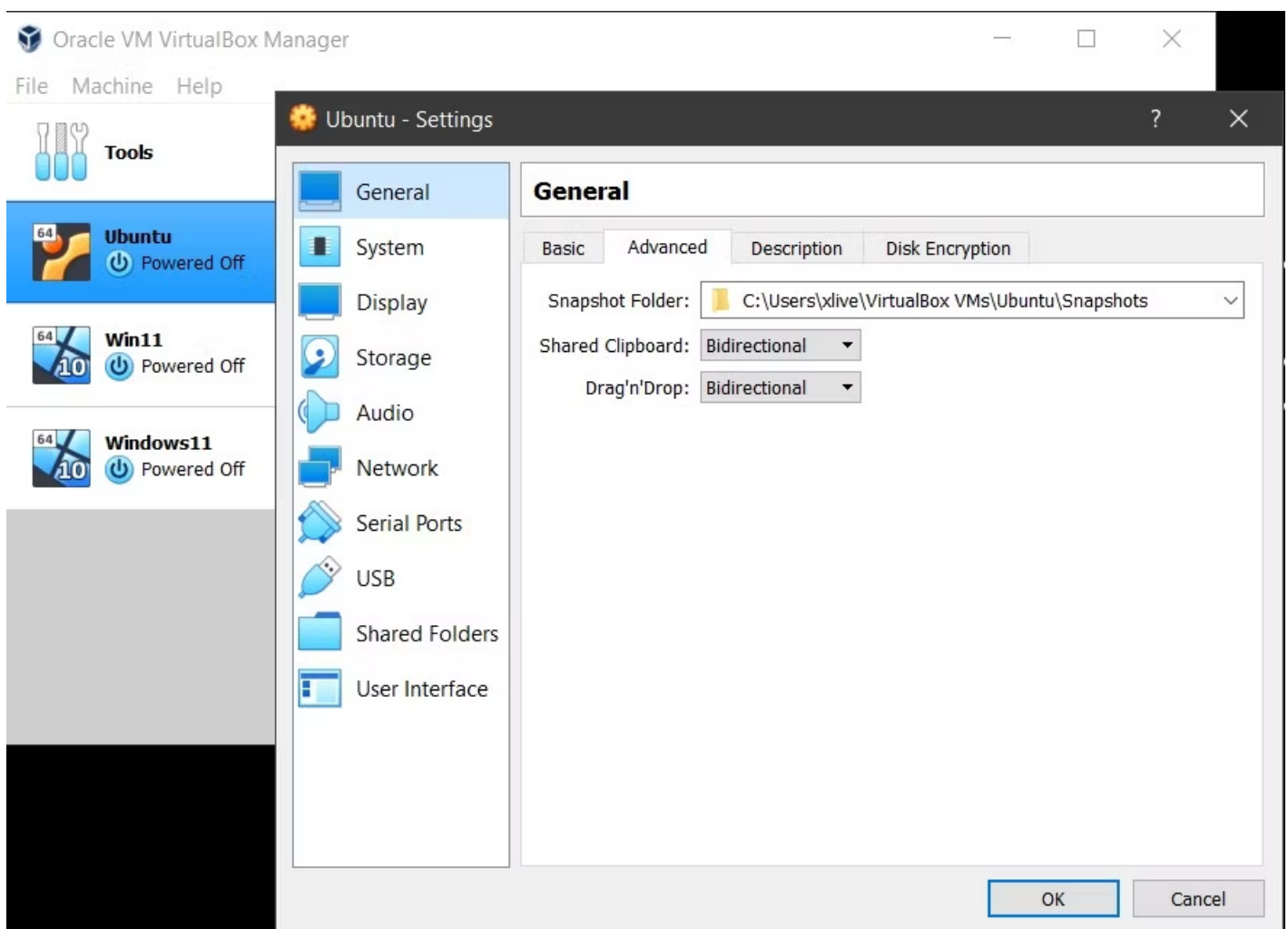
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## Shared Clipboard in VirtualBox

If you're using Oracle VirtualBox, then you can set up a **Shared Clipboard** and **Drag'n'Drop** in just a few quick steps:



1. Select your virtual machine, right-click and select **Settings**.

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2. In the window that pops up, click on **General** and then click on the **Advanced** tab.

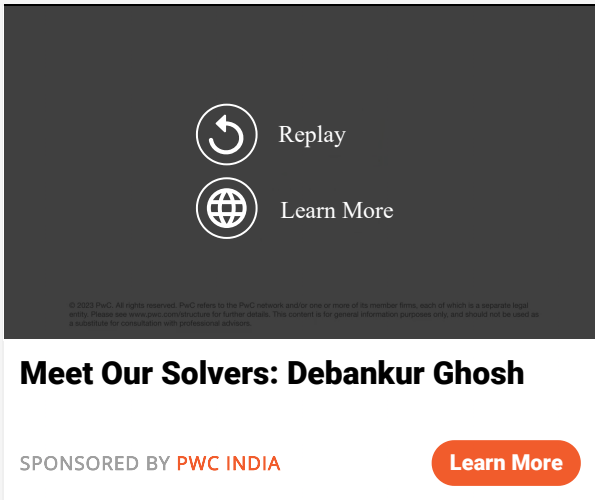
3. You should now see the **Shared Clipboard** and **Drag'n'Drop** dropdown options.

You can choose from **Host to Guest**, **Guest to Host**, and **Bidirectional**. There's also the default option, **Disabled**, which prevents your attempts to copy files to VirtualBox.

Select **Bidirectional** in both drop-down boxes to enable two-way file sharing.

## How to Copy and Paste Data in VMware

Similar to the VirtualBox functionality, you can also set up a shared clipboard or enable drag and drop in VMware. But you might need first to install the VMware Tools package, which brings additional features.



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1. With VMware running, navigate to **VM > Install VMware Tools**. If you haven't already downloaded VMware Tools, instructions for doing so will be given.
2. Enable copy and paste in **VM > Settings > Options**.
3. Select **Guest Isolation**
4. **Here, Enable copy and paste** and **Enable drag and drop**.
5. Confirm with **OK**.

Sharing data between the guest and host operating systems in this way is best suited for

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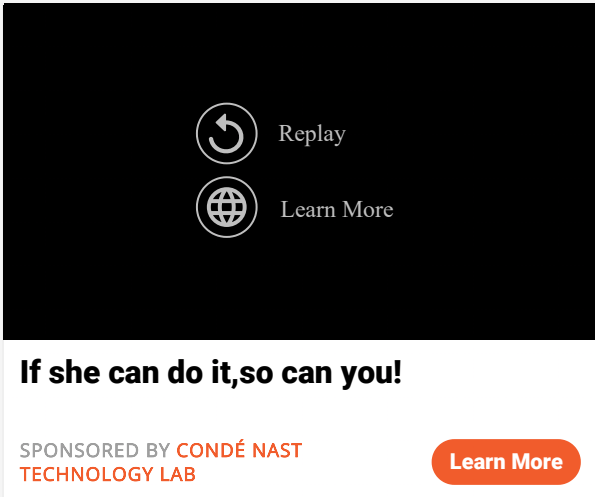
Also share text strings, URLs, that sort of thing. Steer clear of large files, though—you have other options for those.

## 2. Copy Files from VM to the Local Machine On a USB Stick

Using a USB stick to transfer data between two physical machines is a time-honored tradition. Although not ideal, a USB stick can transfer files between the host PC and virtual machine. Let's look at how you can use a USB drive to share data between a virtual machine and your host PC.

### Transfer Files From Host To VirtualBox Using A USB stick

To be able to access USB devices from within VirtualBox, you'll need to enable USB access. For this, you need to download the [VirtualBox Extension Pack](#)



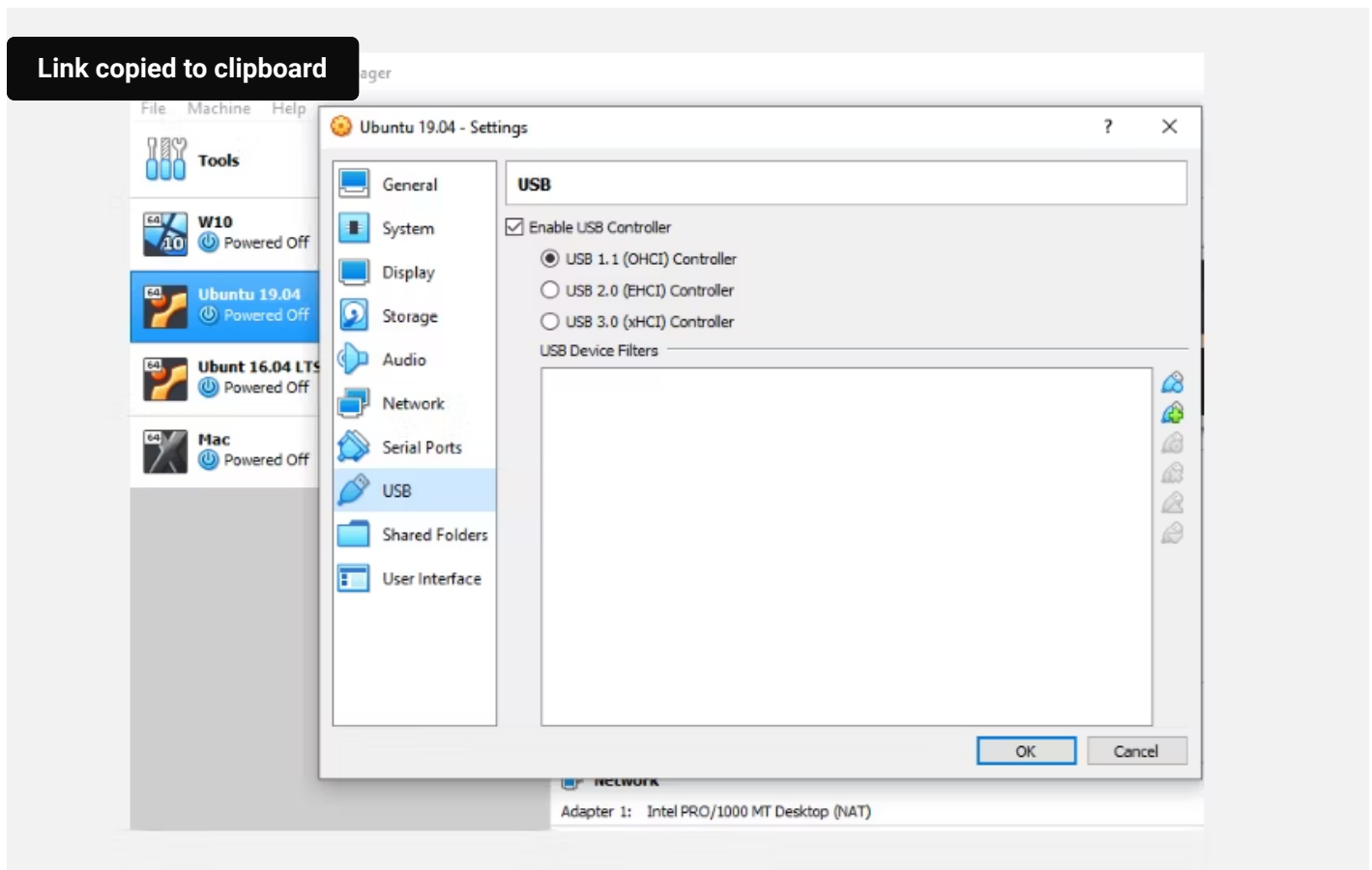
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Once you have downloaded the extension pack:

1. Insert the USB device you wish to use.
2. Launch VirtualBox and click **File > Preferences**, then **Extensions** and click **+**.
3. Browse to the downloaded Extension Pack, click **Open**, then when prompted, **Install**.
4. Follow the prompts to complete the process. You can then check to confirm USB is enabled in **Settings > USB**.
5. Right-click the VM you plan to use and select **Settings > USB**.
6. Click **+** then browse for the USB device, which will be available when you launch the VM. Additional drives can be added in the same way.

You can then use the USB device for file transfer in VirtualBox.

### How To Transfer Files To VMware With A USB Stick

With VMware, when a USB device is connected and the VM is the active window, the device is detected. However, it will not be detected by the host PC in this scenario. For this to happen,

remove the drive, minimize the VM, then reconnect.

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It's simple but can get messy if you forget which operating system the USB stick is connected to.

This option is best if you want to copy large files from VM to host with VMware. Of course, you're limited by the capacity of the USB device, so keep that in mind. Whatever VM software you use, safe ejection of USB devices is recommended on both host and guest virtual machines.

### 3. Create a Shared Folder to Transfer Files from Host to VM

Your third option for copying files from host to guest PC is to set up a network share. This means designating a portion of your PC's hard disk drive as accessible over the local network. With this setup, the VM can then connect to the network and access the drive.

Although physically all on the same computer, this adds higher capacity to your virtual machine data sharing.

#### How to Transfer Files from VirtualBox to Host Using a Shared Folder

You should have already downloaded VirtualBox Guest Additions. This should be installed via **Devices > Install Guest Additions**, where you should browse for the appropriate EXE file. Follow the steps to the end, choosing the default options, then **Finish**.



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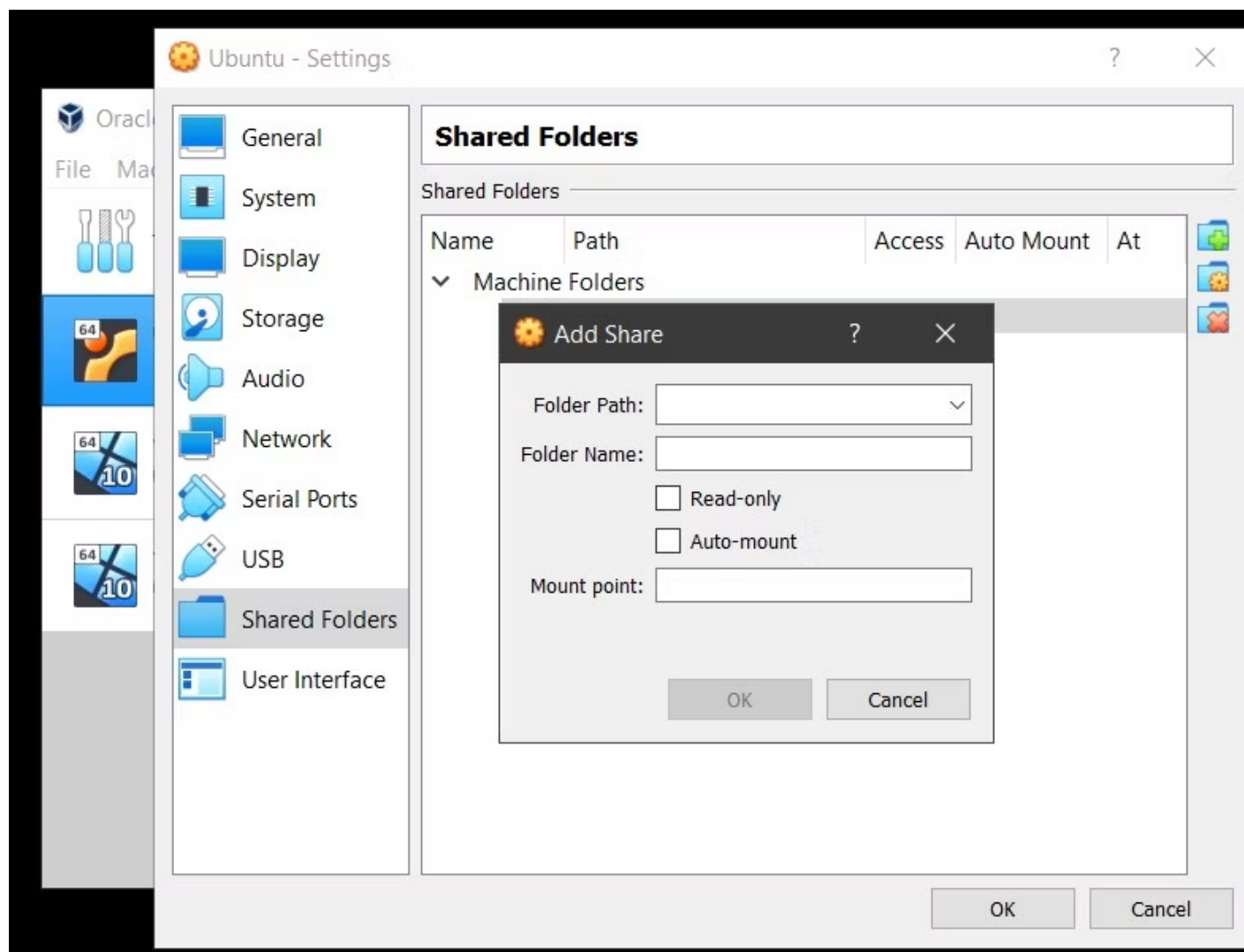
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Launch VirtualBox, then:

### 1. Open **Devices > Shared Folders > Shared Folders Settings**.

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2. Click **+**, then in **Folder Path**, click the arrow and select **Other**.
3. Browse for the folder you plan to share, and click **Select Folder**.
4. In **Add Share**, give the share a name you can use in both operating systems.
5. Check **Auto-mount** and **Make permanent**, then **OK**. From the guest OS, you'll find the share set up in the usual location for network shares. For example, in Windows 10, this will be under **Network Locations** in **Windows Explorer**.

## Share a Network Drive in VMware

The process of creating a shared folder on VMware is straightforward:

1. Within VMware Workstation, choose your virtual machine and click on **Player > Manage > Virtual Machine Settings**
2. Click on **Options > Shared Folders**. You can then choose a sharing option under **Folder Sharing**
3. Click on **Add** to set up your shared folder and follow the on-screen instructions in the **Add Shared Folder Wizard**.
4. You can then type the **Host path** to the directory you want to share and name your folder.
5. Once you click **Next**, you decide the folder access (**Read-only** or **Enable this share**, the latter allows both host and the virtual machine complete access to the shared folder).
6. Click on **Finish** to create your shared folder.



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shared folder under the `/mnt/hgfs` directory.

While this is the most complicated option to set up, you will be able to share larger files effortlessly. These might include installers, disk image files, and hi-res videos.

## Transfer Files Between Virtual Machine and Host

Whether you're using VirtualBox or VMware, sharing data between your physical PC and a virtual machine is surprisingly simple.

If you're not copying files from the host to the guest OS, there's reason to copy data from the VM. This is typically when you're about to destroy the VM and want to back up your data.

Learning how to share data between a VM and host OS will help super-charge your VM use.



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